

JISHNU TEJA DANDAMUDI

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ABOUT

Native of Guntur, Andhra Pradesh and raised in Chennai, Tamil Nadu. I am currently pursuing both a B.Tech (CSE with AI) and B.Sc. (Data Science and Applications) simultaneously, with a strong ambition to join ISRO, Sriharikota and contribute to my country's advancements in space exploration.

EDUCATION

B.Tech in Computer Science Engineering (AI) Amrita Vishwa Vidyapeetham, Coimbatore, Tamil Nadu, India ; CGPA: 7.88/10.00	August 2024 - May 2028
B.Sc in Data Science and Applications Indian Institue of Technology(Madras), Tamil Nadu, India ; CGPA: 9.24/10.00	May 2024 - May 2027
Higher Secondary Class, Class XII Jayanthi Public Senior Secondary School, Tirupur, Tamil Nadu ; 85%	2024
Senior Secondary Class, Class X DAV Boys Senior Secondary School, Tamil Nadu, India ; 87%	2021

SKILLS AND INTERESTS

Interests	Artificial Intelligence, Machine Learning, Deep Learning, Edge Deployment, Quantum AI, Signal Processing, Image & Video Processing
Languages	Python, C, Java, HTML, SQL
Softwares	Ubuntu, Matlab, Canva, Figma, MySQL, scikit-learn, tkinter, pytorch, tensorflow

PROJECTS

1. Neuro-Symbolic Explainable AI for Anomaly Detection in Satellite-IoT Cyber-Physical Systems

November 2025 – January 2026

Internship, Conference Paper

- Proposed a progressive Neuro-Symbolic XAI framework combining CNN–LSTM deep learning with SHAP-based symbolic rule extraction for Satellite-IoT intrusion detection
- Evaluated the model on large-scale heterogeneous datasets (NF-BoT-IoT-V2: 69M+ flows, EDGE-IIoTSET: 7M+ flows), achieving over 95% detection accuracy with high interpretability
- Developed an explainability layer that automatically converts SHAP feature attributions into human-readable IF–THEN security rules for mission-critical satellite CPS environments

2. SOLAR-SAFE: Edge AI-Powered Fault Detection in Photovoltaic Modules using Deep Learning

July 2025 – November 2025

Course Project as a part of curriculum, Conference Paper

- Developed CNN and Involutional Neural Network (INN) models to detect and classify photovoltaic module faults from infrared thermal images
- Trained and evaluated models on the InfraredSolarModules dataset (20,000+ images) using accuracy, precision, recall, and F1-score metrics
- Optimized and deployed models on the Google Coral Edge TPU for low-latency, real-time edge-based PV fault detection

3. CNN Meets INN: Efficient DNA Motif Prediction using Deep Learning

July 2025 – November 2025

Course Project as a part of curriculum, Conference Paper

- Developed hybrid CNN and Involutional Neural Network (INN) architectures for accurate DNA motif prediction across multiple genomic categories
- Trained and evaluated models on the JASPAR benchmark dataset, achieving up to 99.8% accuracy with strong generalization across species
- Performed a comparative analysis of CNN and INN models, highlighting efficiency–accuracy trade-offs and improved performance of INNs for specific taxa

4. Prune Smart, Train Sharp: ADMM-Powered CNN and INN Models for Image Classification

July 2025 – November 2025

Course Project as a part of curriculum, Conference Paper

- Designed and trained CNN and Involutional Neural Network (INN) models for image classification on the CIFAR-10 benchmark
- Applied ADMM-based structured pruning to CNN and INN architectures, achieving higher accuracy and model compression
- Evaluated and compared baseline and ADMM-optimized models, demonstrating improved accuracy–efficiency trade-offs for resource-constrained deployment

5. Voltage Stability Enhancement in Microgrids using ANN-Based Droop Control

January 2025 – March 2025 Course Project as a part of curriculum, Conference Paper - 2nd Best Paper Award

- Proposed an ANN-based adaptive droop control strategy integrated with state-space modeling to enhance voltage and frequency stability in microgrids
- Developed and simulated dynamic droop and ANN controllers in Python, demonstrating reduced voltage deviation and improved frequency regulation under varying load conditions
- Performed comparative analysis with conventional and optimized droop control methods, validating superior stability and robustness of the ANN-based approach

6. Bridging Vision and Language using CLIP–ResNet Framework for Visual Question Answering

January 2025 – March 2025

Course Project as a part of curriculum, Conference Paper

- Proposed a multimodal Visual Question Answering (VQA) framework integrating CLIP for textual understanding and ResNet-18 for visual feature extraction
- Implemented image–text feature fusion through deep learning and trained the model on the DAQUAR dataset, achieving an accuracy of 95.4
- Conducted comprehensive performance analysis using accuracy and weighted precision metrics, demonstrating improved semantic alignment over conventional BERT–ResNet models

7. Detection of Power System Anomalies using a Fusion of Machine Learning and Deep Learning

January 2025 – March 2025

Course Project as a part of curriculum, Conference Paper

- Proposed a hybrid anomaly detection framework combining LSTM Autoencoders and Isolation Forest for PMU-based power system monitoring
- Addressed class imbalance using SMOTE and evaluated the model on real-world labelled PMU datasets covering faults, disturbances, and cyber-induced anomalies
- Achieved balanced anomaly detection performance with 88% precision by fusing temporal and statistical anomaly scores for improved grid resilience

8. Airline Reservation System using Python Tkinter and Java

January 2025 – March 2025

Course Project as a part of curriculum

- Developed a functional Airline Reservation System integrating Python–Tkinter for the GUI and Java modules for backend operations, enabling seat booking, cancellation, and passenger information management.
- Designed modular user interface workflows and implemented core reservation logic with structured file-based data handling to ensure smooth and consistent system operation.
- Organized and managed multi-language project components (Python, Java, HTML), demonstrating full-stack development skills and implementing basic validation, ticket processing, and system interaction flow.

9. Dynamic Feature Learning with Involution and Convolution for Predominant Instrument Recognition in Polyphonic Music

November 2024

Journal Paper

- Contributed to a published journal work presenting a hybrid deep-learning framework combining Convolutional Neural Networks (CNN) and Involution Neural Networks (INN) for polyphonic predominant instrument recognition.
- Implemented Mel-spectrogram-based feature extraction and designed CNN + INN architectures to capture global frequency patterns and localized spatial features from complex audio mixtures.
- Developed a soft-voting ensemble that integrated complementary CNN and INN predictions, achieving state-of-the-art performance while significantly reducing computational complexity.
- Experimentally validated the model on the IRMAS dataset, demonstrating improvements over existing baseline approaches in terms of micro/macro F1-scores and overall robustness.

10. Spatio-Channel Complementary Learning for Polyphonic Music Instrument Recognition

November 2024

Conference Paper

- Developed an ensemble deep-learning framework combining Convolutional Neural Networks (CNN) and Involution Neural Networks (INN) for predominant instrument recognition in polyphonic music.
- Designed and implemented complete audio preprocessing pipeline, including Mel-spectrogram generation, model training, and performance benchmarking on the IRMAS dataset.
- Achieved significant improvements over state-of-the-art baseline models, demonstrating higher micro and macro F1-scores with reduced computational complexity.

11. Design and Development of Arduino-Based Autonomous Firefighting Robot

August 2024 – November 2024

Course Project as a part of curriculum, Conference Paper

- Designed and developed an Arduino-powered autonomous firefighting robot integrating flame sensors, ultrasonic sensors, a servo mechanism, and a water-pump-based suppression system.
- Conducted extensive software (TinkerCAD) and hardware prototyping, performance testing, and quantitative analysis; achieved improved detection accuracy and reduced extinguishing time across varying distances.

12. Design and Implementation of HACK CPU and Synchronous Down Counter

July 2024 – November 2024

Course Project as a part of curriculum

- Designed and implemented a complete 16-bit HACK CPU architecture including ALU, registers (A, D, PC), memory units, and control logic using HDL and hardware simulation tools.
- Built and verified fundamental logic components such as adders, multiplexers, 16-bit arithmetic units, memory blocks (RAM8, RAM64), and the integrated CPU datapath.
- Developed a synchronous MOD-10 down counter ($9 \rightarrow 0$) using JK flip-flops, K-maps, state diagrams, and Boolean minimization for reliable edge-triggered counting.
- Performed simulation, validation, and integration testing of all chips, demonstrating correct CPU instruction execution and stable counter operation under synchronous clocking.

13. Railway Reservation System using Python Tkinter

2023-2024

CSC Project in School

- Developed a complete GUI-based Railway Reservation System using Python (Tkinter) with MySQL integration, enabling train search, ticket booking, PNR enquiry, cancellation, and booking history management.
- Designed separate Admin and User modules, including features such as train creation, route configuration, seat allocation, reservation charts, and user-side ticket management workflows.
- Implemented modular database connectivity, booking logic, and interface components, demonstrating the ability to build multi-module desktop applications with persistent data handling.

RESEARCH PUBLICATION

1. Rupa Kandula, **Jishnu Teja Dandamudi**, Rayappa David Amar Rayappa, Rama Muni Reddy Yanamala and K. Krishna Prakasha

Explainable AI for Smart Grid Intelligence: A Comprehensive Review of Techniques, Applications and Future Directions.

Energy and AI, Elsevier - 100820 (2026).

2. **Jishnu Teja Dandamudi**, Rupa Kandula, Rayappa David Amar Raj, Rama Muni Reddy Yanamala, Christian Napoli and Cristian Randieri

Deep Reinforcement Learning for Wireless Networks: Architectures, Algorithms, and Emerging Applications.

Taylor & Francis Group, (2026).

3. Rupa Kandula, **Jishnu Teja Dandamudi**, Rayappa David Amar Raj, Rama Muni Reddy Yanamala, Christian Napoli and Cristian Randieri

Artificial Intelligence-empowered 6G Wireless Networks: Architectures, Techniques, and Future Directions.

Taylor & Francis Group, (2026).

4. **Jishnu Teja Dandamudi**, Rupa Kandula, Rayappa David Amar Raj, Rama Muni Reddy Yanamala, Christian Napoli and Cristian Randieri

Artificial Intelligence for Sustainable Wireless Network Management: Models, Strategies, and Optimization.

Taylor & Francis Group, (2026).

5. Rupa Kandula, **Jishnu Teja Dandamudi**, Rayappa David Amar Raj, Rama Muni Reddy Yanamala, Christian Napoli and Cristian Randieri

Principles and Innovations in Sustainable Wireless Communication Systems for Environmental Efficiency.

Taylor & Francis Group, (2026).

6. **Jishnu Teja Dandamudi**, Rupa Kandula, Rayappa David Amar Rayappa, Rama Muni Reddy Yanamala and K. Krishna Prakasha

Explainable AI for wind energy systems: State-of-the-art techniques, challenges, and future directions. *Energy Conversion and Management: X*, Elsevier - 101277 (2025).

7. **Jishnu Teja Dandamudi**, Yogendra Chhetri

Prune Smart, Train Sharp: ADMM-Powered CNN and INN Models for Image Classification.

IEEE, (2026).

8. **Jishnu Teja Dandamudi**, Rupa Kandula

Voltage Stability Enhancement in Microgrids: An ANN-based Droop Control Approach.

Taylor & Francis Group, (2026).

9. **Jishnu Teja Dandamudi**, Yogendra Chhetri, Prajeesh C.B and Abhishek S

Solar-Safe: An Edge AI - Powered Fault Detection in PhotoVoltaic Modules Using Deep Learning Models. *IEEE*, (2026).

10. Lekshmi C.R., & Jishnu Teja Dandamudi
Spatio-Channel Complementary Learning for Polyphonic Music Instrument Recognition.
IEEE, (2026).

11. Jishnu Teja Dandamudi
Creating a GUI-Based Train Ticket Booking System in Python. *Opast Publishers*, (2025).

12. Lekshmi C.R., & Jishnu Teja Dandamudi
Dynamic Feature Learning with Involution and Convolution for Predominant Instrument Recognition in Polyphonic Music. *Circuits, Systems, and Signal Processing*, 1–18 (2025).

Please click the link below to view the details of all 20 research papers submitted to date, including those that have been accepted.

PROPOSALS

Enhanced Neural Network for Classification of Hyperspectral Imagery using Ensemble models	
Respond Basket, ISRO	2024
Solid Rocket Motor Design and Safety	
Dare to Dream 5.0, DRDO	2024

INTERNSHIPS

Research Intern, CYSTAR, IIT Madras Hybrid Internship.	Jan, 2026 - Present
Intern, NRSC, ISRO Deep Guage Radar: High Resolution Rainfall Estimation through Deep Learning Physical Internship.	May, 2026 - ISRO, 2026
Intern, SDSC SHAR, ISRO NeuroSight: Edge-Based Human Recognition & Detection System Physical Internship.	Apr, 2026 - May, 2026
Research Intern, Ministry of Defence, New Delhi Worked under the guidance of Wing Commander Yogendra Chhetri on the topic "Neuro-Symbolic Explainable AI for Cyber-Physical Satellite IoT Systems: Real-Time Fault Diagnosis and Autonomous Decision-Making". Virtual Internship.	Nov, 2025 - Jan, 2026
Tech Intern, NetElixir Digital Solutions (P) Ltd., Hyderabad I was asked to develop a small part of it which is done to a level of 95% at present. Physical Internship.	May, 2025 - June, 2025

POSITION OF RESPONSIBILITY

Amrita Vishwa Vidyapeetham, Coimbatore	
· Research and Development Wing Member, ELITE Club	Dec 2025 – Present
· Research and Development Volunteer, Tensor AI Club	Jul 2025 – Jul 2026
· Deputy Cultural Secretary, Chinmayi House	Jan 2026 – Present

Indian Institute of Technology (Madras)

- Research and Development Core Team, RaSoR Society
- Technical Community Head, Nilgiri House
- Regional Coordinator (Tamil Nadu), Nilgiri House
- Research and Development Core Team, AVASYA Society

Aug 2025 – Present
Aug 2025 – Feb 2026
Jul 2025 – Jan 2026
Mar 2025 – Oct 2025

IEEE Student Branch

- Vice Chair and Founder, IEEE Computer Society

Aug 2025 – Present

ACHIEVEMENTS

- 1st Place in Debate - **NIRC of IITM** October 2025
- 1st Place in Pythonic Coding Competition 4.0- **PARADOX, IITM** May 2025
- 2nd Best Paper Award in a Conference - **ICRAIC2IT** May 2025
- 1st Python Sprint 2.0 - **NIRC of IITM** December 2024
- 2nd Prize in Chatbot Development in Hackathon - **MAGNIFICA2k23** August 2023
- 1st Prize in Chatbot Development - **DAVBGPM** August 2022

RECOMMENDATIONS

- **Dr. Rama Muni Reddy Yanamala**
Assistant Professor, Department of Electronics and Communication Engineering,
Indian Institute of Information Technology Design and Manufacturing (IIITD&M), Kancheepuram
Ph: 9849440181
- **Yogendra Chhetri**
Wing Commander - Directorate of Amphibious & Space Operations,
Integrated Defence Staff, Ministry of Defence, New Delhi
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- **Dr. David Amar Raj Rayappa**
Assistant Professor, Amrita School of Artificial Intelligence,
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Ph: 9441385577
- **Dr. Lekshmi C R**
Assistant Professor, Amrita School of Artificial Intelligence,
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- **Dr. Premjith B**
Assistant Professor, Amrita School of Artificial Intelligence,
Amrita Vishwa Vidyapeetham, Coimbatore
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DECLARATION

I hereby declare that all the details furnished above are true to the best of my knowledge and belief.